



Epidemiology of Oral Cancer



Cancer is a broad term. It describes the disease that results when cellular changes cause the uncontrolled growth and division of cells.

Cancer may be regarded as a group of diseases characterized by:

- Abnormal growth of cells.
- Ability to invade adjacent tissues and even distant organs.
- The eventual death of the affected patient.

Types of cancers

- 1) Carcinomas
- 2) Sarcomas
- 3) Lymphomas
- 4) Leukaemias

- Oral Cancer is one of the ten most common cancers in the world.
- Oral cancer term includes cancers of (lip, tongue, buccal mucosa, floor of mouth and pharynx).
- Oral cancer is classically described as an indurated, ulcerated lump or sore that may or may not be painful and is often associated with cervical lymph adenopathy.
- 90 to 95% of all oral cancers are squamous cell carcinomas.

Etiology of oral cancer

A)Host Factors:

1.Age: Incidence increases with age and 85 % of cases are found in those aged 50 and above. Older age shows increase incidence in carcinoma whereas the younger age shows increase in sarcoma.

2.Race: Whites develop lip melanoma more frequently than the blacks. Certain odontogenic tumours are more common in black races.

3.Gender: commonly Males are more prone than females. Cancers of tongue and buccal mucosa are more common in males.

4. Genetic factors: Discovery of oncogenes introduced a time in which it is possible to identify genetic elements involved in the initiation and progression of malignant disease.

5.Occupation: Textile workers show an increase in oral cancer. Leather workers show an increase in cancer of buccal mucosa.

6. Immunity: Kaposi sarcoma is more common in AIDS patients.

7. Social class: There is a definite relationship between socioeconomic status and frequency of cancer. Low-income groups show increase in cancer of oral cavity.

8. Customs and habits: Smoking increases the incidence of cancer.

B)Agent Factors:

I. Biological: (a) Virus (HIV, HSV), (b) Fungus (Candida).

II. Chemical: dyes, nickel, aromatic amines, chromium.

III. Mechanical: Sharp tooth, any other source of chronic irritation like ill-fitting dentures, chronic sores, etc.

IV. Nutritional agents: Precarcinogens in food (saccharin, aflatoxin), increased consumption of fat, deficiency of folic acid, protein deficiency, decrease in zinc, vegetables, vitamins E and C.

c)Environmental Factors:

I. Water contaminants: It includes some organic pollutants like chloroform.

II. Air pollution: Air pollution caused by the release of a number of gases from the automobiles and factories, e.g. carbon dioxide.

III. Geographic variations: In Netherlands, buccal mucosa is most commonly affected and is more often seen in males.

IV. Solar heat: Prolonged exposure to sunlight causes melanoma.

V. Industrialization: The release of various toxins by the industries contaminates water and air, which may lead to cancer.

Levels of prevention for oral cancer

A)Primary Prevention

1. Avoid tobacco and alcohol use.
2. Avoid smoking.
3. Avoid exposure to sun.
4. Ensure a healthy diet free from vitamin and nutritional deficiency.

B)Secondary Prevention

Patients whose cancer is detected at an early stage generally have much longer survival times than those with late-stage disease.

- Screening of high-risk groups
- Biopsy: any suspicious oral mucosal lesion including any non-healing ulcer .

Levels of prevention for oral cancer

c)Tertiary Prevention

- Surgery, radiotherapy, and chemotherapy.
- In order to stop the recurrence and spread of oral cancers, dentists and other health specialists should work together to provide multi-disciplinary support for patients.
- Treated patients may still have dental needs which dentists should monitor to maintain life quality.
- Prevention of caries by topical fluoride application, dietary advice.
- Management of a dry mouth, and prosthetic rehabilitation following surgery and radiation therapy.

Rehabilitation after Oral Cancer

Rehabilitation may vary from person-to-person depending on the type of oral cancer treatment, and the location and extent of the cancer.

Rehabilitation may include:

- Dietary counselling: Many patients recovering from oral cancer surgery has difficult eating, so it is often recommended that they eat small meals consisting of soft, moist foods.

- Surgery: Some patients may benefit from reconstructive or plastic surgery to restore the bones or tissues of the mouth, returning a more normal appearance.
- Prosthesis: If reconstructive or plastic surgery is not an option, patients may get benefit from dental or facial-part prosthesis to restore a more normal appearance. Special training may be needed to learn to use a prosthetic device. speech
- Speech therapy: If a patient experiences difficulty in speaking following oral cancer treatment,



**THANKS
FOR LISTENING**

